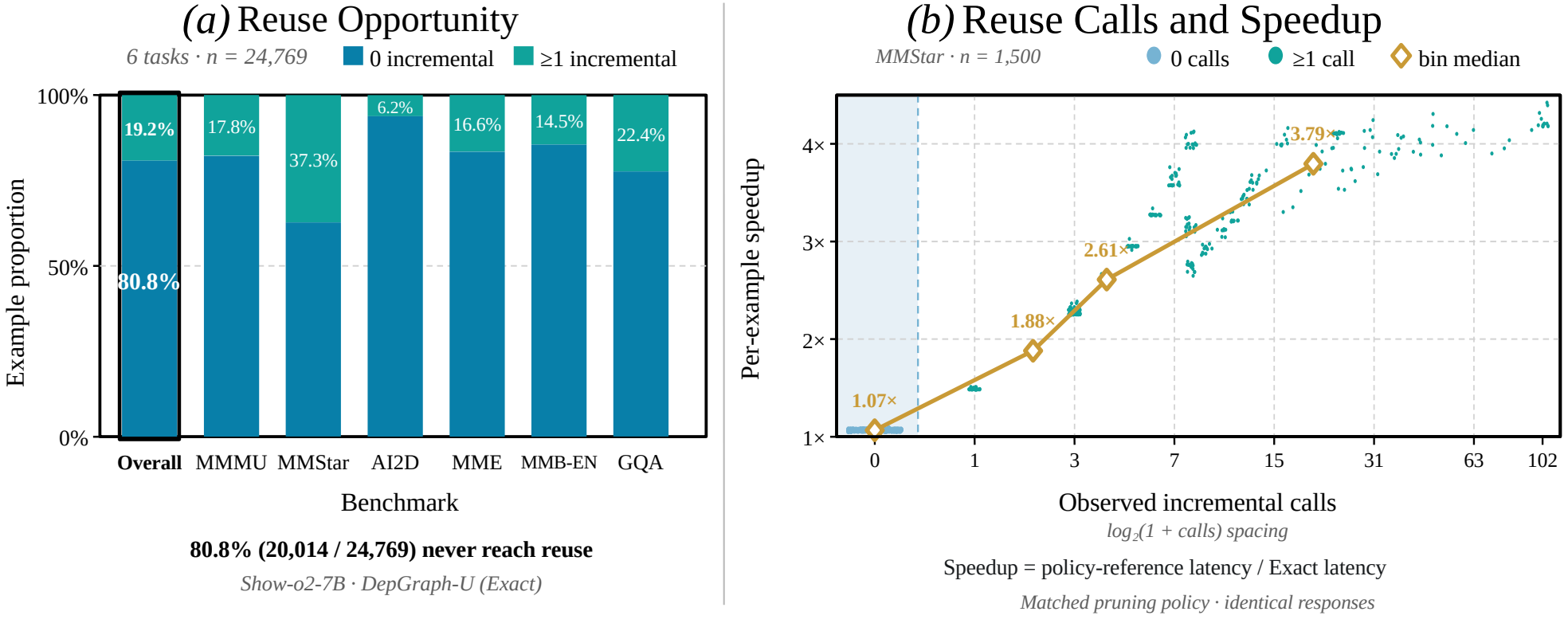

(a) Reuse Opportunity
6 tasks · n = 24,769
0 incremental
≥1 incremental
100%
50%
0%
Example proportion
19.2%
80.8%
17.8%
37.3%
6.2%
16.6%
14.5%
22.4%
Overall
MMMU
MMStar
AI2D
MME
MMB-EN
GQA
Benchmark
80.8% (20,014 / 24,769) never reach reuse
Show-o2-7B · DepGraph-U (Exact)
(b) Reuse Calls and Speedup
MMStar · n = 1,500
0 calls
≥1 call
bin median
Per-example speedup
4×
3×
2×
1×
1.07×
1.88×
2.61×
3.79×
0
1
3
7
15
31
63
102
Observed incremental calls
log2(1 + calls) spacing
Speedup = policy-reference latency / Exact latency
Matched pruning policy · identical responses